Time Series Analysis of Gold ETF Prices

DATS 6313 – Time Series Analysis
Final Term Project

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4. Abstract

This project presents a comprehensive time series analysis of Gold ETF Close Prices, aiming to uncover temporal patterns and develop predictive models for future price forecasting. Utilizing a multivariate dataset sourced from Kaggle, which includes key financial indicators such as S&P 500, NASDAQ, CPI, GDP, and various commodity prices, the study focuses on modelling the gold price over time.

Initial exploratory analysis reveals that the gold price series exhibits significant non-stationarity, influenced by strong trend and moderate seasonality. Statistical tests such as the Augmented Dickey-Fuller (ADF) and KPSS confirm the need for transformation. A thorough decomposition using the STL method helps isolate the trend and seasonal components, further supporting model selection.

Various forecasting models are explored, including Base Models, Holt-Winters Exponential Smoothing, ARMA, ARIMA, and Box-Jenkins methodologies. Regression analysis is also conducted using macroeconomic variables, incorporating feature selection techniques like PCA and VIF to address multicollinearity. Baseline models (naïve, drift, average) are used for comparison against more complex methods.

The final selected model is determined based on diagnostic tests, error metrics (MSE, RMSE, AIC, BIC), and forecasting accuracy on the test set. The analysis concludes with h-step ahead forecasting, and (if implemented) a web-based dashboard is proposed to visualize model predictions interactively.

This study highlights the effectiveness of time series modelling in financial forecasting and provides valuable insights into the behaviour and predictability of gold prices, supporting data-driven investment decision-making.

5. Introduction

Time series analysis is a crucial statistical technique used to model and predict sequential data that evolves over time. In financial markets, time series methods are extensively employed to analyse asset prices, identify patterns, and make data-driven forecasts. These techniques leverage temporal dependencies, trends, seasonality, and cyclical behaviour to understand underlying dynamics in time-stamped data.

This project focuses on the time series analysis and forecasting of Gold ETF Close Prices using historical financial data. Gold is widely recognized as a hedge against inflation and market volatility, making it a critical asset in diversified portfolios. By analysing its price behaviour over time, we aim to build predictive models that can provide valuable insights for investors and analysts.

The analysis begins with preprocessing and exploring the dataset, which contains several economic and market variables including equity indices (S&P 500, NASDAQ), commodity prices (oil, silver, platinum), and macroeconomic indicators (CPI, GDP). The selected dependent variable is the Gold ETF Close Price, and the goal is to model its behaviour using both traditional time series models and machine learning-inspired regression approaches.

The project follows a structured approach:

Step 6: Dataset description, preprocessing, and exploratory analysis using plots, correlation matrices, and autocorrelation tools.

Step 7: Stationarity checks using ADF and KPSS tests, along with rolling statistics and necessary transformations like differencing.

Step 8: Decomposition of the series into trend, seasonal, and residual components using STL (Seasonal-Trend decomposition).

Step 9: Forecasting with the Holt-Winters method.

Steps 10–12: Feature selection using PCA/VIF and regression modelling to assess linear relationships.

Step 13–15: Development of ARMA, ARIMA, SARIMA, and Box-Jenkins models with order selection using the GPAC method, and LM algorithm.

Step 16: Residual diagnostics and validation tests to ensure model robustness.

Steps 17–19: Forecasting future gold prices using selected models, evaluating h-step predictions, and comparing performance across all modelling techniques.

The final objective is to select the most accurate model for forecasting Gold ETF Close Prices, supported by statistical evaluation and visualization. Additionally, a dashboard may be developed to present model outputs in an interactive format (bonus).

This comprehensive process demonstrates the power of time series analysis in uncovering insights from financial data and building models capable of future prediction with statistical confidence.

6- Description of the dataset

6 a.

```
Missing values in each column:
sp500 close        0
nasdaq close       0
CPI                0
GDP                0
silver close       0
oil close          0
platinum close     0
palladium close    0
gold open          0
gold high          0
gold low           0
gold close         0
gold volume        0
dtype: int64
```

Fig. 1

6 d.

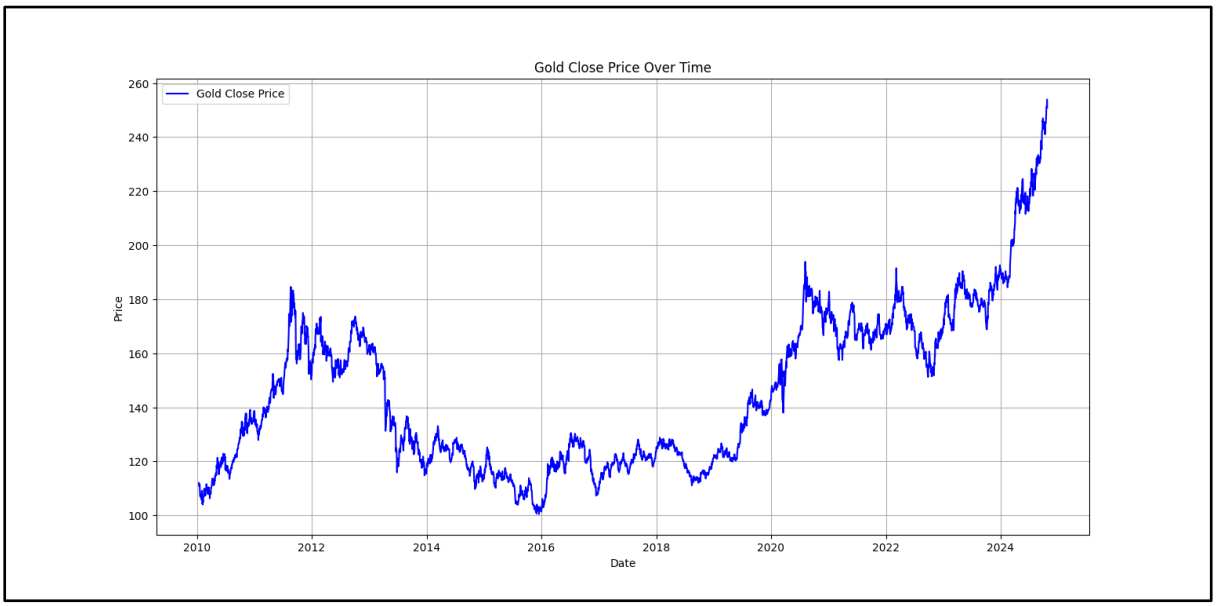

Fig. 2

Plot of the Dependent Variable (Gold Close Price vs Time)

Observations:

- The Gold Close Price shows a clear upward trend over time, especially from around 2020 onwards.
- The time series is non-stationary, evidenced by a steadily increasing mean and fluctuations in variance over time.
- There are cyclical fluctuations that suggest potential seasonality or long-term cycles in the gold market, possibly influenced by macroeconomic factors like inflation, geopolitical tensions, and market speculation.
- The recent rise in prices looks exponential, which may indicate volatility and momentum-driven price movements.

6 e.

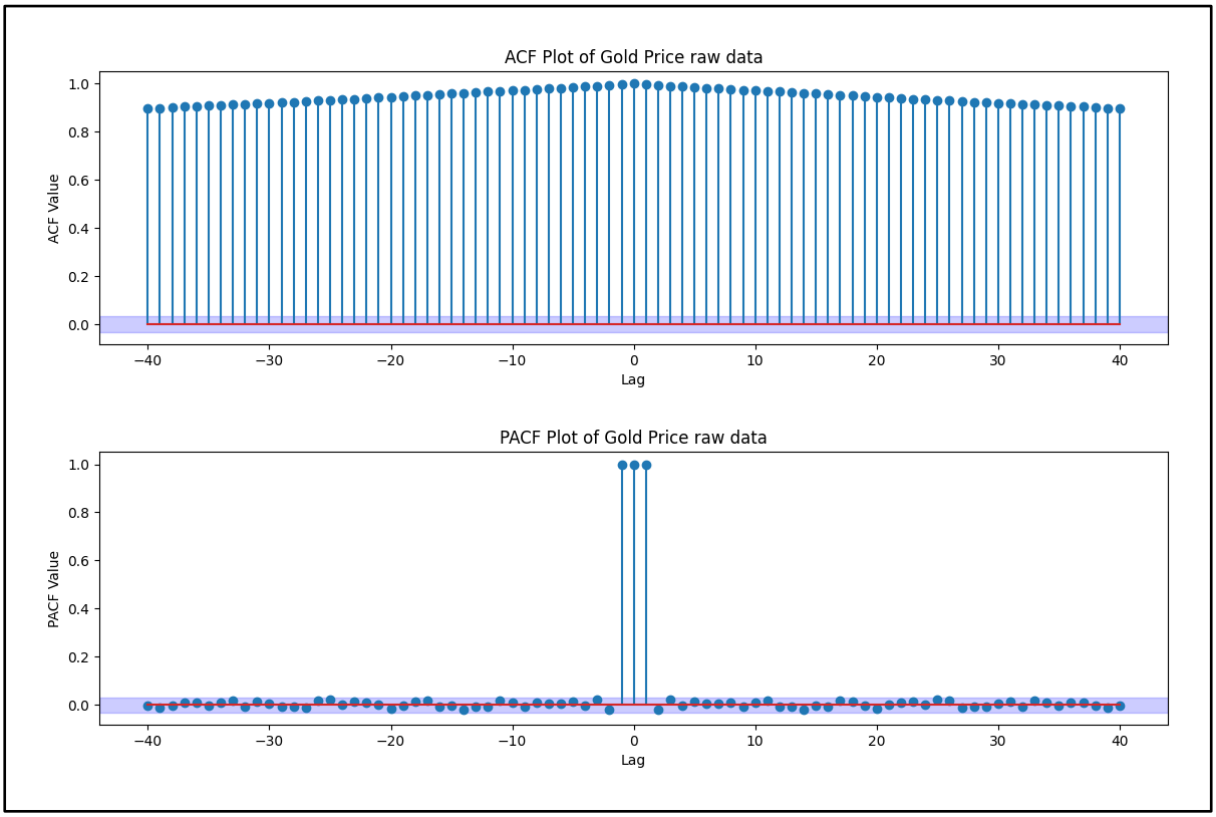

Fig. 3

ACF and PACF of the Raw Data

Observations:

- ACF (Autocorrelation Function):
  - Extremely high autocorrelation at all lags, starting close to 0.95 and declining only slightly as lag increases.
  - This behaviour indicates a strong trend and non-stationarity, as the time series retains memory of its past values.
  - Suggests that simple models like AR might not be sufficient without differencing.
- PACF (Partial Autocorrelation Function):
  - Significant spike at lag 1, followed by values near zero at all higher lags.
  - This implies that the time series might be autoregressive of order 1 (AR (1)) once made stationary.
  - The sharp drop after lag 1 in PACF supports the idea of a short-memory process once the trend is removed.

6 f.

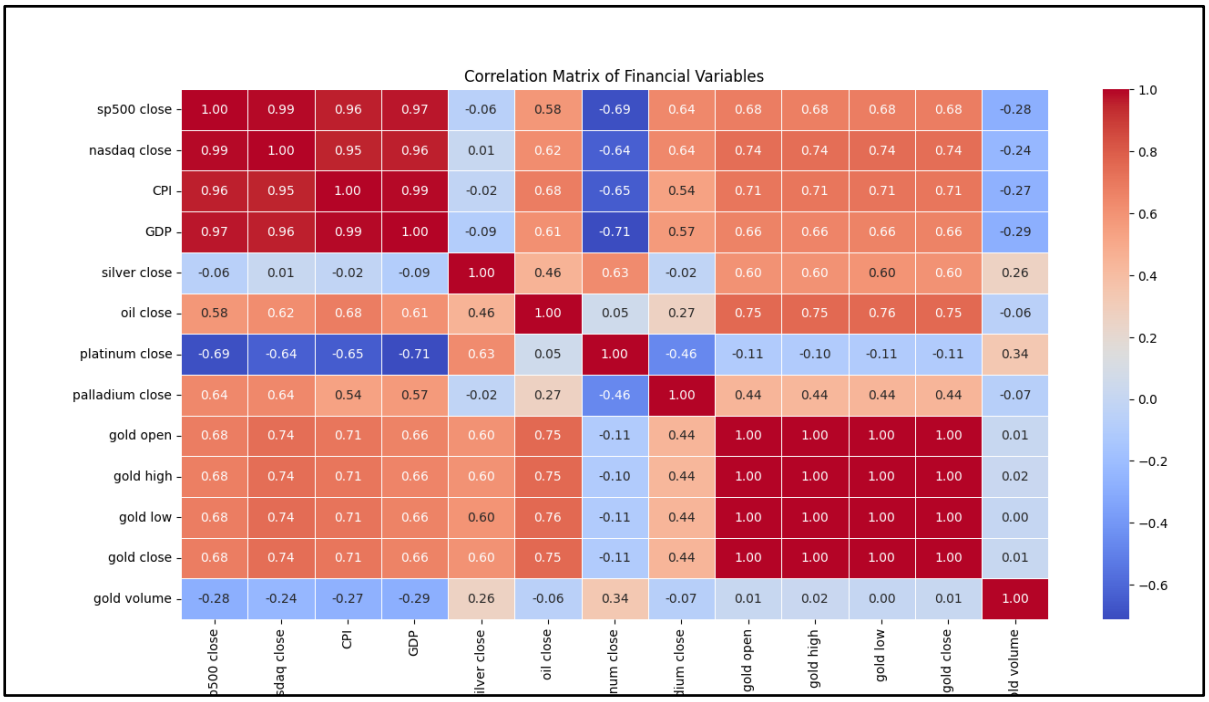

Fig. 4

Correlation Matrix of Financial Variables

Observations:

- Strong positive correlation between:
  - sp500 close, Nasdaq close, CPI, and GDP (correlation > 0.95).
  - These variables are likely collinear and should be considered for dimensionality reduction or feature selection.
- gold close is highly correlated with:
  - gold open, gold high, and gold low (correlation ≈ 1.00) - indicating redundancy.
- gold volume has very low or negative correlation with most variables, suggesting it may not contribute much explanatory power.
- platinum close has a negative correlation with major indices and CPI, suggesting it may move differently from traditional equity and inflation indicators.

6 g.

```
Training set shape: (3123, 13)
Testing set shape: (781, 13)
```

Fig. 5

7- Stationarity

```
Calculating rolling mean and variance for Gold Price...
Initial Mean: 112.0300, Final Mean: 145.4278
Initial Var : 0.0000, Final Var : 874.8286
ADF Test:
  ADF Statistic: 0.3768
  p-value: 0.9806
  Critical Values: {'1%': -3.4320265580345004, '5%': -2.8622808115385583, '10%': -2.567164342737072}
  Conclusion: Non-stationary (fail to reject H0)


KPSS Test:
  KPSS Statistic: 4.2427
  p-value: 0.0100
  Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}
  Conclusion: Non-stationary (reject H0)
```

Fig. 6

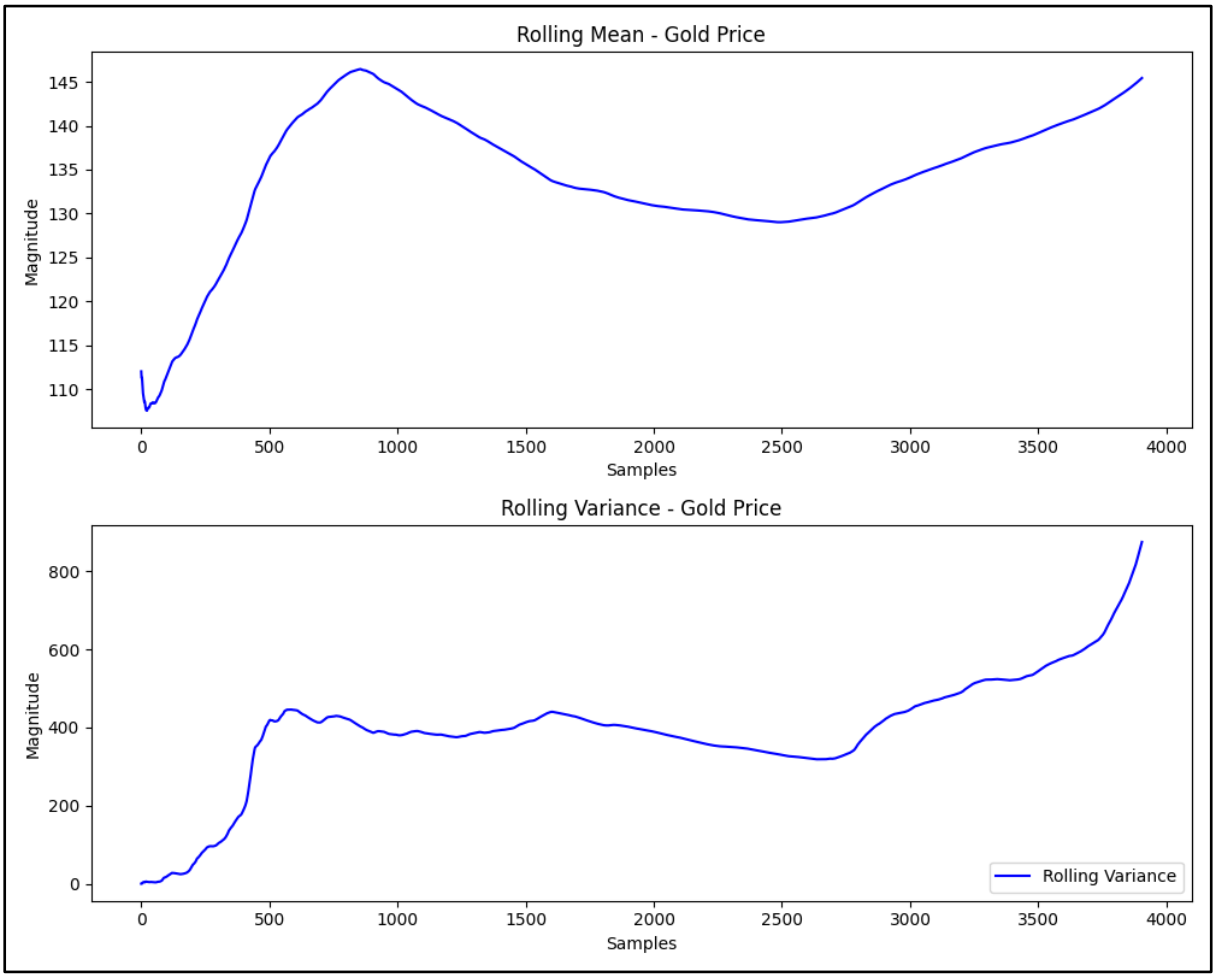

Fig. 7

The Gold Close data was non-stationary.

1st order differencing performed.

```
Initial Mean: -1.1700, Final Mean: 0.0356
Initial Var : 0.0000, Final Var : 2.0530


ADF Test:
  ADF Statistic: -62.6211
  p-value: 0.0000
  Critical Values: {'1%': -3.43202698798399, '5%': -2.862281001464468, '10%': -2.5671644438460373}
  Conclusion: Stationary (reject H0)


KPSS Test:
  KPSS Statistic: 0.3148
  p-value: 0.1000
  Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}
  Conclusion: Stationary (fail to reject H0)
```

Fig. 8

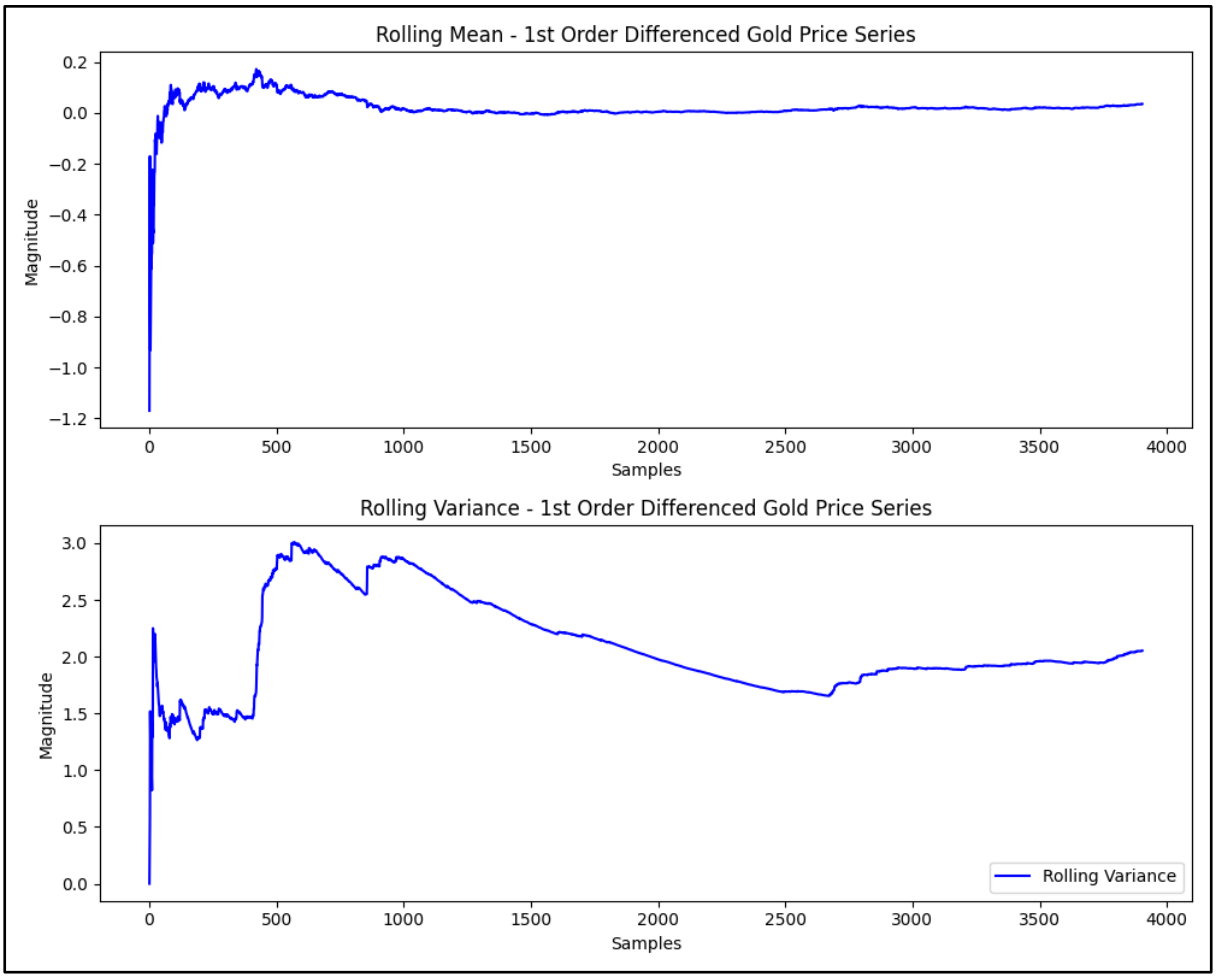

Fig. 9

The Gold Close data is transformed into stationary.

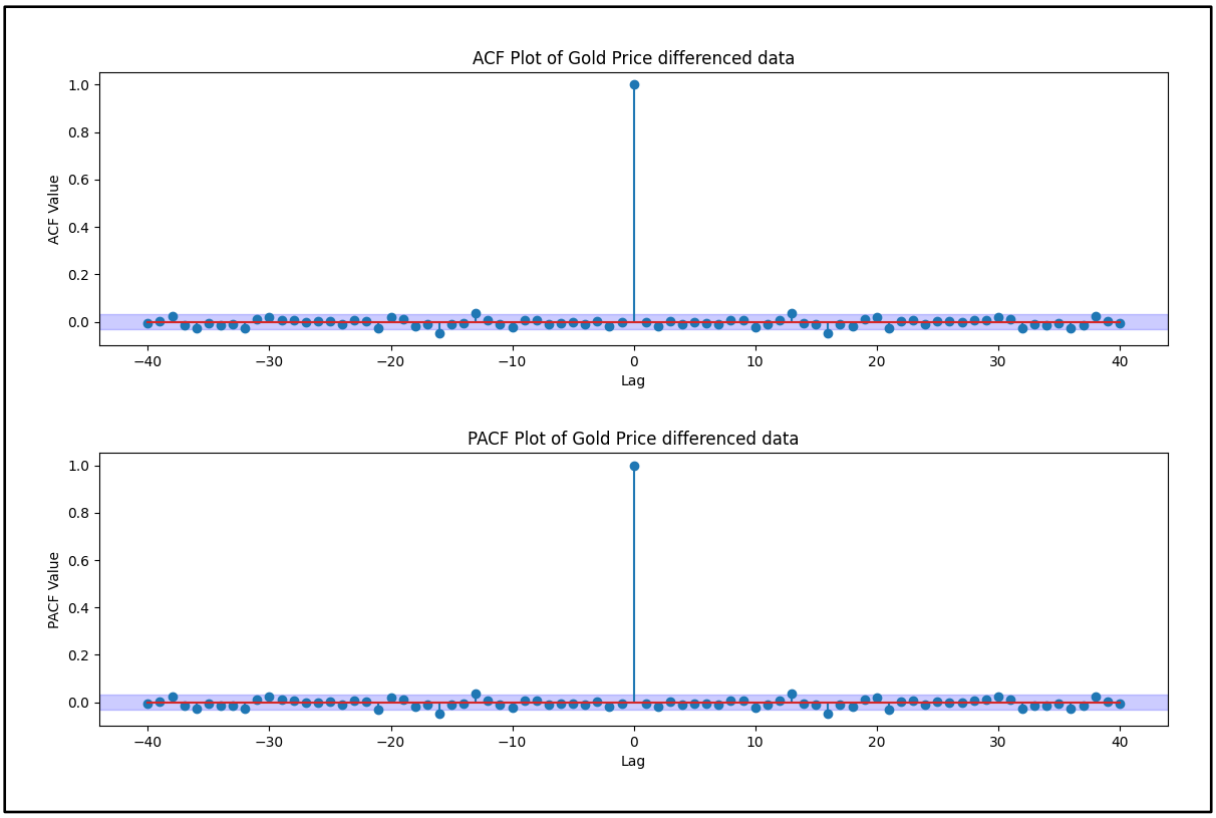

Fig. 10

ACF Plot of the differenced Gold close data

8- Time series Decomposition

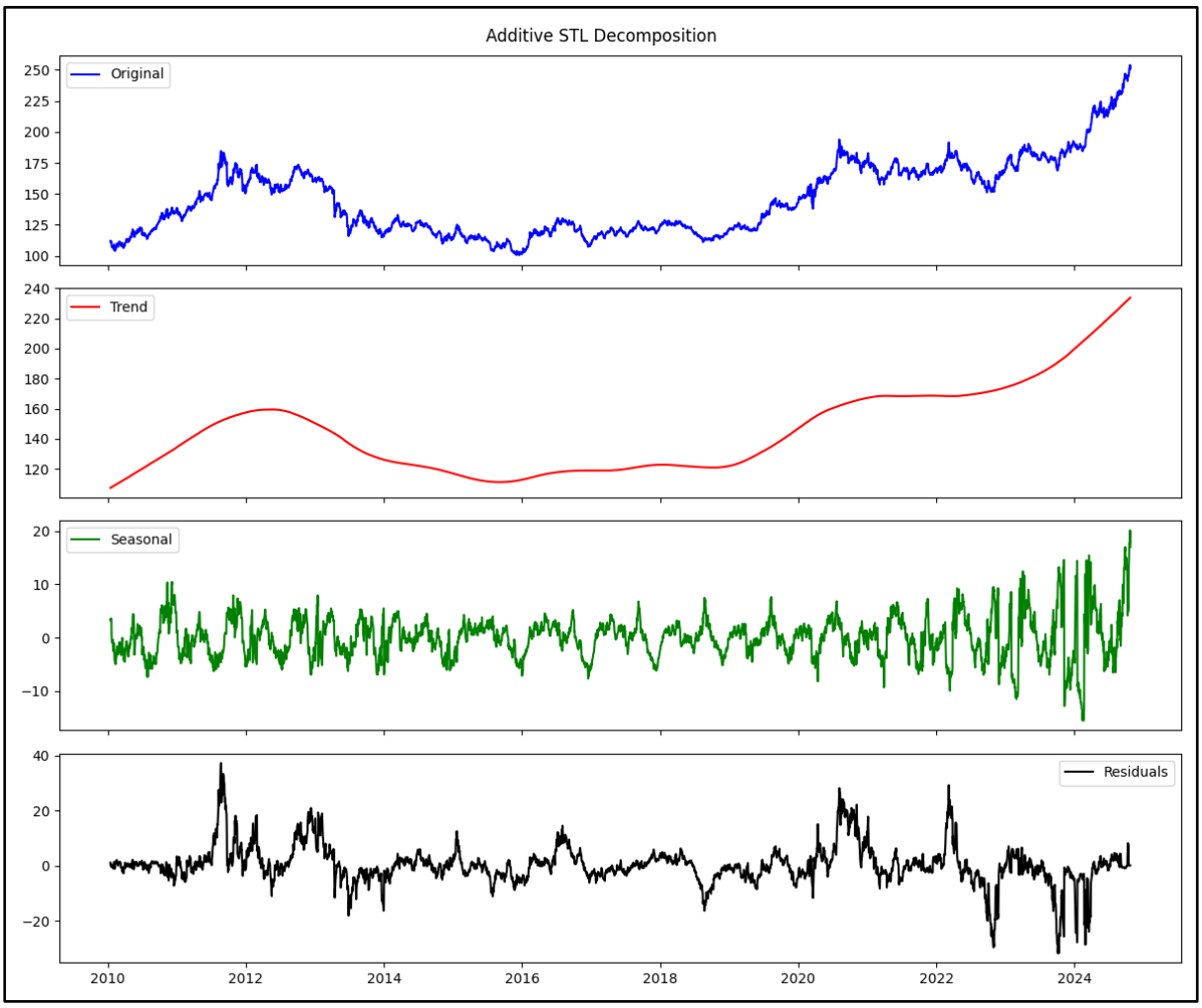

Fig. 11

```
Additive Decomposition:
  Strength of Trend: 0.9406
  Strength of Seasonality: 0.0580
```

```
Residual Variance (Additive): 50.6573
```

Fig. 12

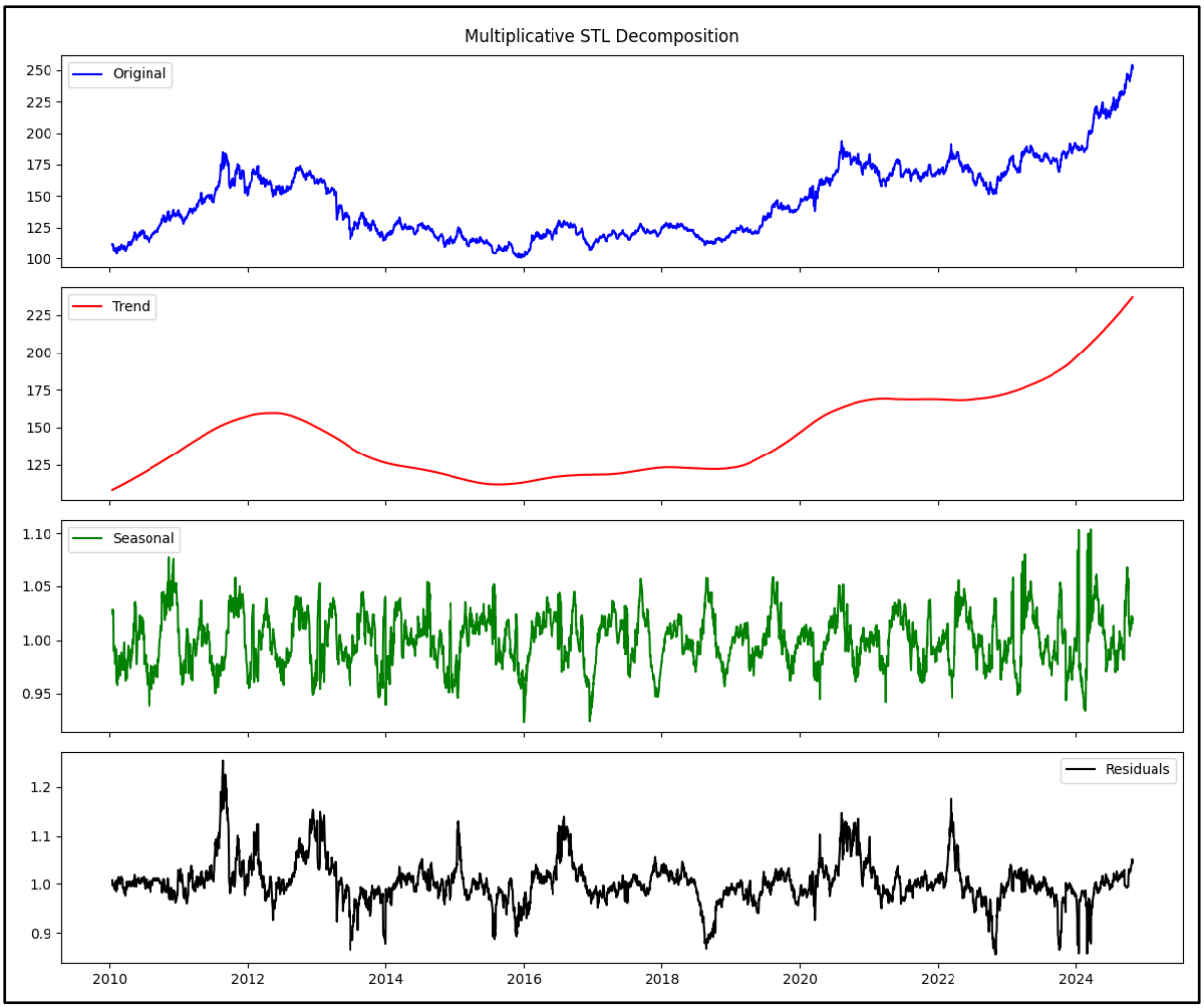

Fig. 13

```
Multiplicative Decomposition:
  Strength of Trend: 1.0000
  Strength of Seasonality: 0.1202
```

```
Residual Variance (Multiplicative): 0.0020
```

Fig. 14

Performed both additive and multiplicative STL decompositions on the gold price time series using a seasonal period of 252 (trading days in a year) to extract trend and seasonality components. The additive decomposition showed a strong trend (0.9406) but very weak seasonality (0.0580), with moderate residual variance (50.6573). In contrast, the multiplicative decomposition captured the trend nearly perfectly (1.0000), showed slightly stronger seasonality (0.1202), and had significantly lower residual variance (0.0020), indicating a better fit.

Visual inspection of the decomposed components confirmed that the multiplicative model more accurately captured the proportional growth and seasonal structure of the series. Based on the strength metrics and residual behaviour, the multiplicative decomposition is more appropriate for this dataset.

9- Holt-Winters Method

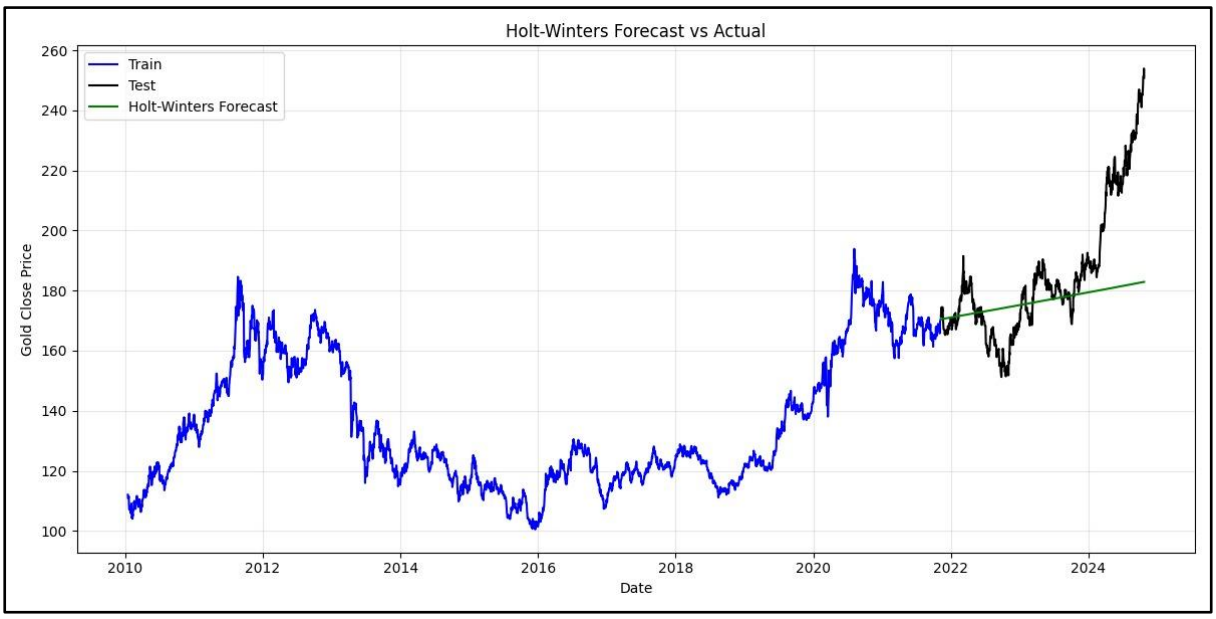

Fig. 15

```
Holt-Winters Forecast Evaluation:
  Mean Absolute Error (MAE): 14.2446
  Mean Squared Error (MSE): 455.7504
  Root Mean Squared Error (RMSE): 21.3483
```

Fig. 16

To model and forecast the gold closing prices, the Holt-Winters exponential smoothing method with a multiplicative trend component was applied. The model was trained on 80% of the dataset and evaluated on the remaining 20%.

Upon forecasting, the model achieved a Mean Absolute Error (MAE) of 14.24, a Mean Squared Error (MSE) of 455.75, and a Root Mean Squared Error (RMSE) of 21.34.

The forecast captures the general upward trend in gold prices but tends to underestimate sharp fluctuations in recent years. This indicates that while the Holt-Winters method performs well for identifying consistent seasonal and trend behaviour, it may fall short in modelling the volatility and abrupt movements common in financial time series like gold.

10- Feature selection/dimensionality reduction

```
VIF Analysis:
            feature          VIF
0       sp500 close   174.386016
1      nasdaq close   145.709646
2               CPI   307.708503
3               GDP   304.039147
4      silver close     7.661793
5         oil close    13.197141
6    platinum close    16.816221
7   palladium close     2.170254
8         gold open  2950.673911
9         gold high  3454.833226
10         gold low  3195.754881
11      gold volume     2.279893

Condition Number: 193.50

Singular Values (SVD):
[167.08180269 104.38058535  59.20822583  47.3285381   38.46565206
  19.99651952  16.61468327  10.69309107   3.58128578   2.53717362
   0.94582786   0.86345931]

Final selected features:
['sp500 close', 'nasdaq close', 'CPI', 'GDP', 'silver close', 'oil close', 'platinum close', 'palladium close', 'gold volume']
```

Fig. 17

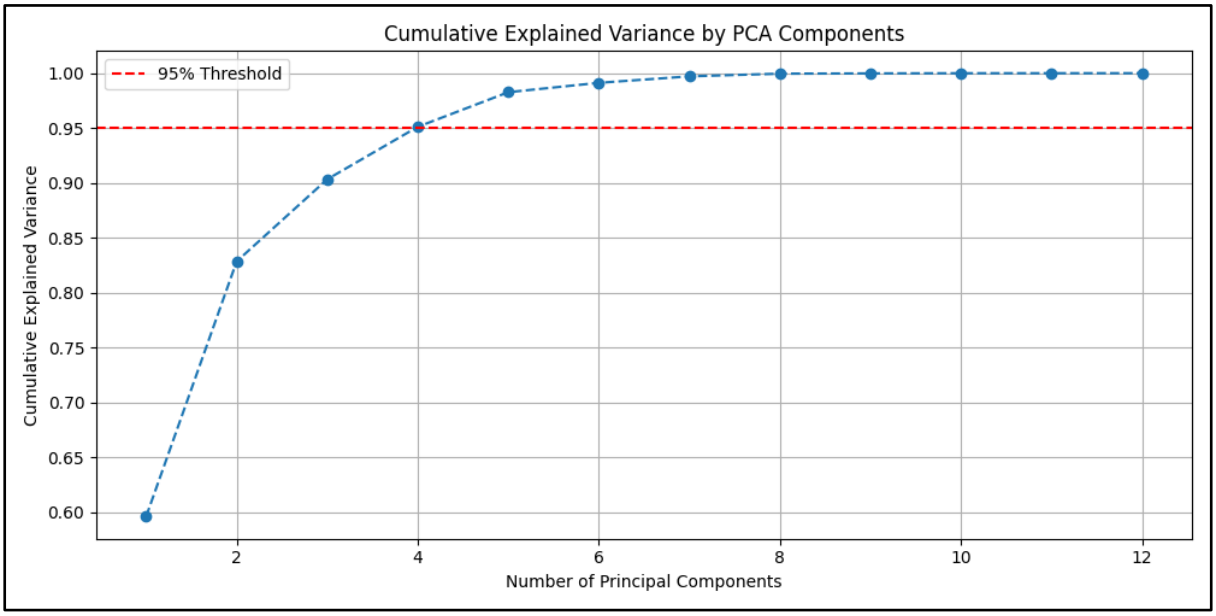

Fig. 18

To address multicollinearity and improve model interpretability, feature selection was performed using PCA, VIF analysis, SVD, condition number, and backward stepwise regression.

The condition number of the standardized feature matrix was 193.5, which, while not extremely high, suggests moderate multicollinearity in the dataset. A condition number above 30 generally indicates potential issues, and values nearing 200 confirm the need for careful feature selection.

PCA showed that over 95% of the variance is explained by the first 4 components, suggesting redundancy in some features. VIF analysis revealed that "gold open", "gold high", and "gold low" had extremely high VIF values (all above 2900), indicating strong multicollinearity. These features are highly correlated with "gold close" and with each other, making them redundant. Therefore, they were eliminated to improve model stability.

Finally, backward stepwise regression dropped features with high p-values, retaining only the most significant predictors. The final selected features were:

["sp500 close", "nasdaq close", "CPI", "GDP", "silver close", "oil close", "platinum close", "palladium close", "gold volume"].

This combination of methods ensured a well-conditioned, interpretable set of predictors for the regression model.

11- Base Models

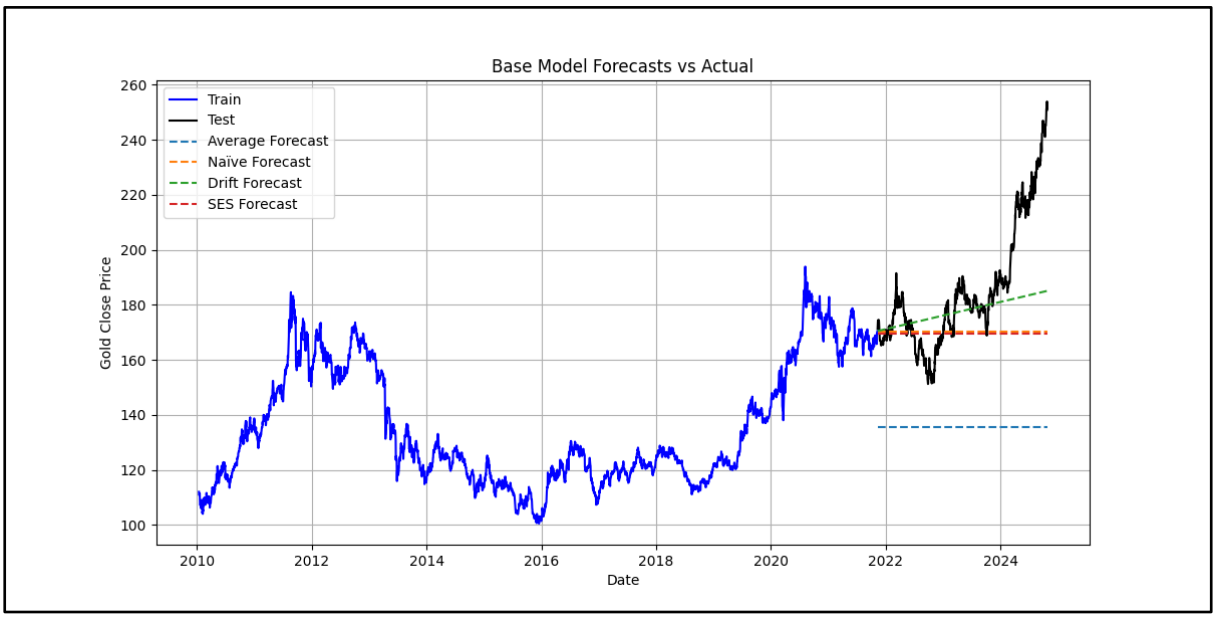

Fig. 19

```
Base Model RMSEs:
  Average Model      : 54.5309
  Naïve Model        : 26.7726
  Drift Model        : 20.5011
  SES Model          : 27.3327
```

Fig. 20

- Average Model RMSE = 54.53

  Performs the worst; it uses a constant flat forecast (mean of training), which clearly fails to capture the upward trend in recent years.

- Naïve Model RMSE = 26.77

  Uses last training value repeatedly; better than average but still lags behind dynamic changes in test data.

- Drift Model RMSE = 20.50

  Best performing among the base models. It extrapolates the trend from training data, which aligns well with the test set's continued rise.

- SES Model RMSE = 27.33

  Similar to naïve model in performance, though slightly worse here. Likely because SES with constant smoothing can't adapt fast to trend shifts.

The forecast lines match this performance:

- Drift forecast rises and tracks the test set better.
- Others (especially average) remain flat and diverge quickly.

Among the base models evaluated, the Drift model performed the best with the lowest RMSE of 20.50, effectively capturing the upward trend in gold prices. In contrast, the Average model performed poorly due to its flat forecasts. The Naïve and SES models offered moderate accuracy but failed to adapt to the trend. Overall, the Drift model is most suitable for this dataset.

12- Multiple Linear Regression Model (OLS)

12 a.

An MLR model using 9 predictors and achieved an R^2 of 0.9234, indicating that over 92% of the variance in gold prices is explained by the model. One-step-ahead forecasts closely followed actual values during the test period, visually confirmed in your prediction plot.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             gold close   R-squared:                       0.923
Model:                            OLS   Adj. R-squared:                  0.923
Method:                 Least Squares   F-statistic:                     4170.
Date:                Fri, 02 May 2025   Prob (F-statistic):               0.00
Time:                        17:54:30   Log-Likelihood:                -10040.
No. Observations:                3123   AIC:                         2.010e+04
Df Residuals:                    3113   BIC:                         2.016e+04
Df Model:                           9
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const            -412.7146     13.283    -31.071      0.000    -438.759    -386.670
sp500 close        -0.7025      0.017    -42.504      0.000      -0.735      -0.670
nasdaq close        0.5677      0.014     39.665      0.000       0.540       0.596
CPI                 2.9217      0.073     40.193      0.000       2.779       3.064
GDP                -0.0071      0.000    -16.385      0.000      -0.008      -0.006
silver close        1.6486      0.038     43.284      0.000       1.574       1.723
oil close          -0.3919      0.028    -14.010      0.000      -0.447      -0.337
platinum close      0.2097      0.019     10.770      0.000       0.172       0.248
palladium close     0.0731      0.006     11.395      0.000       0.061       0.086
gold volume     -6.722e-08   1.91e-08     -3.516      0.000   -1.05e-07   -2.97e-08
==============================================================================
Omnibus:                      207.087   Durbin-Watson:                   0.041
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              276.620
>>>
```

Fig. 21

12 b.

Hypothesis Testing (F-test and t-tests):

- F-test (Prob(F-statistic) ≈ 0): The overall model is statistically significant.
- t-tests: All predictors have p-values well below 0.05, indicating that each contributes significantly to predicting the gold close price.

12 c.

```
CV Mean MSE: 257.5113
```

Fig. 22

The mean MSE = 257.51 from rolling-window cross-validation confirms model stability and generalization performance across different time splits. This adds robustness to in-sample fit.

12 d.

```
MSE: 679.3961
RMSE: 26.0652
AIC: 20099.12
BIC: 20159.59
R²: 0.9234
Adjusted R²: 0.9232
```

Fig. 23

Evaluation Metrics:

- MSE = 679.40, RMSE = 26.07: Error is moderate and acceptable given real-world financial data.
- AIC = 20099.12, BIC = 20159.59: These values help in comparing models, with lower values being better.
- Adjusted R^2 = 0.9232 reinforces the model's explanatory power after adjusting for the number of predictors.

12 e.

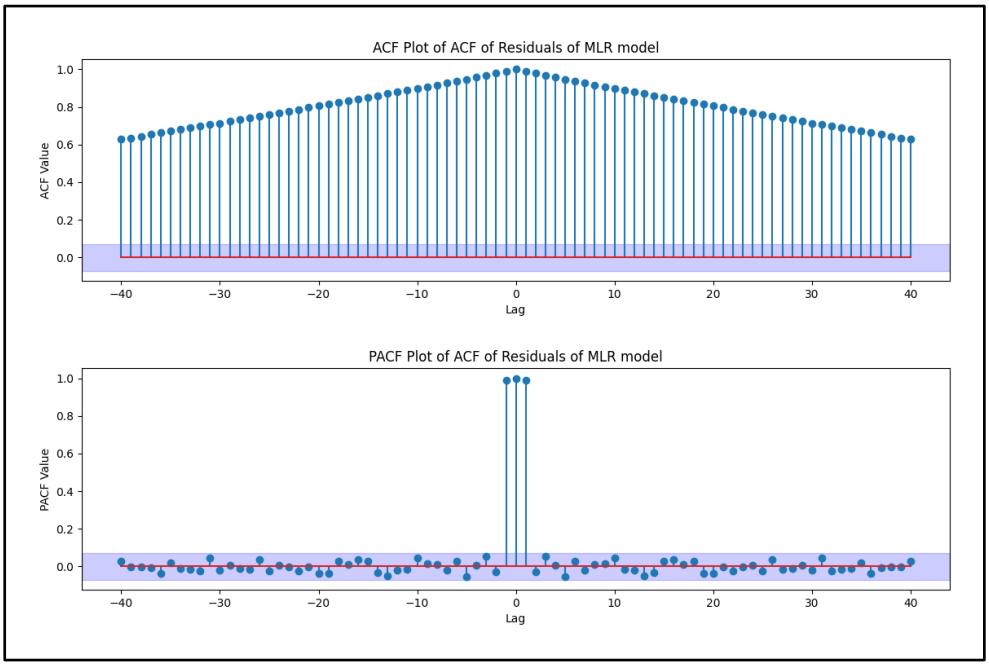

Fig. 24

The ACF plot shows significant autocorrelation in residuals, violating the independence assumption. This implies that some time-dependent patterns are not fully captured by the MLR model.

12 f.

```
Ljung-Box Q-Statistic (lag=20): 12737.0199
p-value: 0.0
```

Fig. 25

With Q = 12737.02 and p < 0.001, the residuals are clearly autocorrelated. This statistically confirms the ACF finding and signals that model assumptions are not fully satisfied.

12 g.

```
Residual Mean: -18.9591
Residual Variance: 320.3587
```

Fig. 26

- Mean = -18.96: Not centred around 0 — a potential bias.
- Variance = 320.36: Indicates a sizable spread in residual errors, suggesting heteroskedasticity or omitted temporal structure.

12 h.

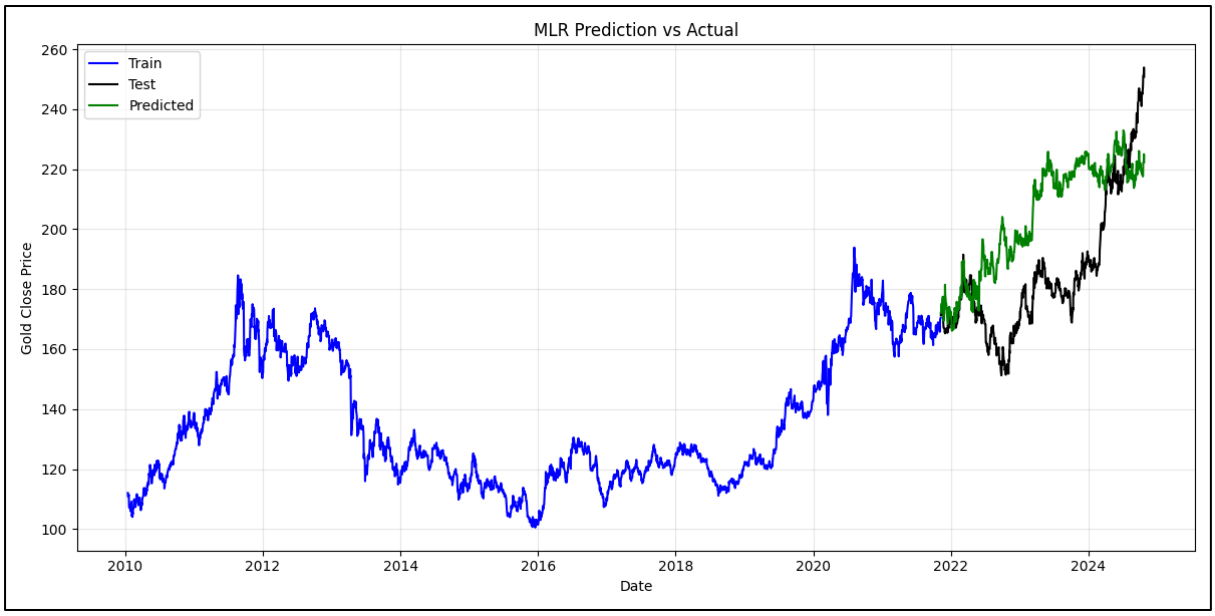

Fig. 27

Plot shows that the model fits the training set well and tracks the test data trends, although it starts to diverge slightly due to unresolved autocorrelation in residuals.

The MLR model provides a strong predictive baseline with high explanatory power ($R^2 = 0.9234$) and statistically significant predictors. It performs well on both training and test sets, as seen in the visual forecast alignment and cross-validated MSE. However, residual diagnostics show clear autocorrelation and a non-zero mean, indicating violations of independence and unbiasedness assumptions. This suggests the model could benefit from time-series enhancements (e.g., ARMAX) to capture temporal dependencies. Nonetheless, as a foundational linear model, it effectively establishes relationships between macroeconomic indicators and gold prices.

13- ARMA-ARIMA-SARIMA/Multiplicative model

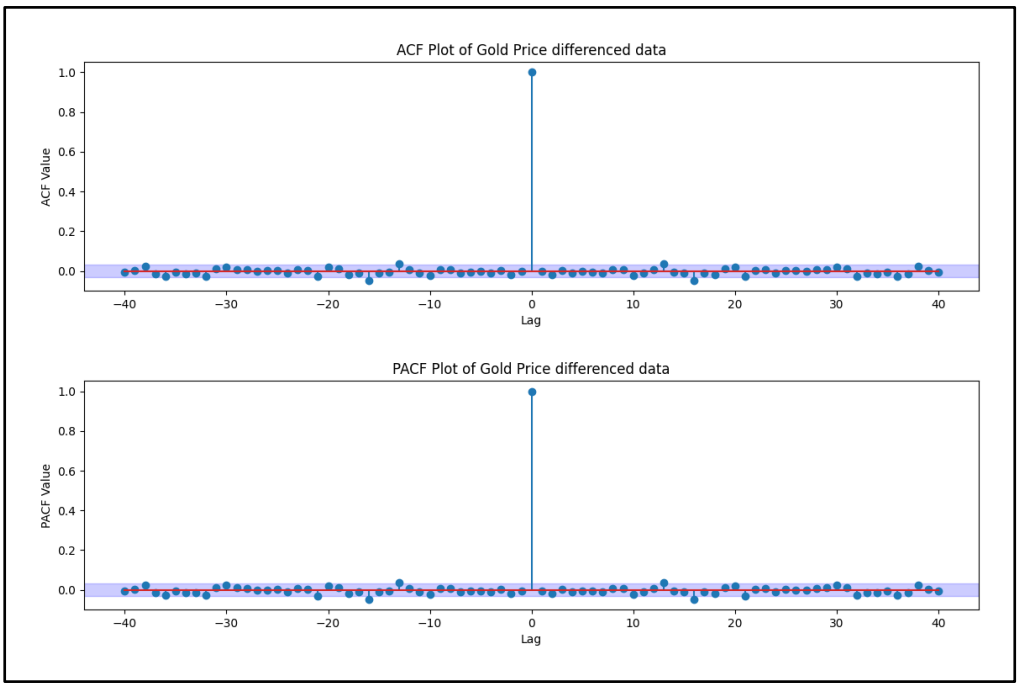

Fig. 28

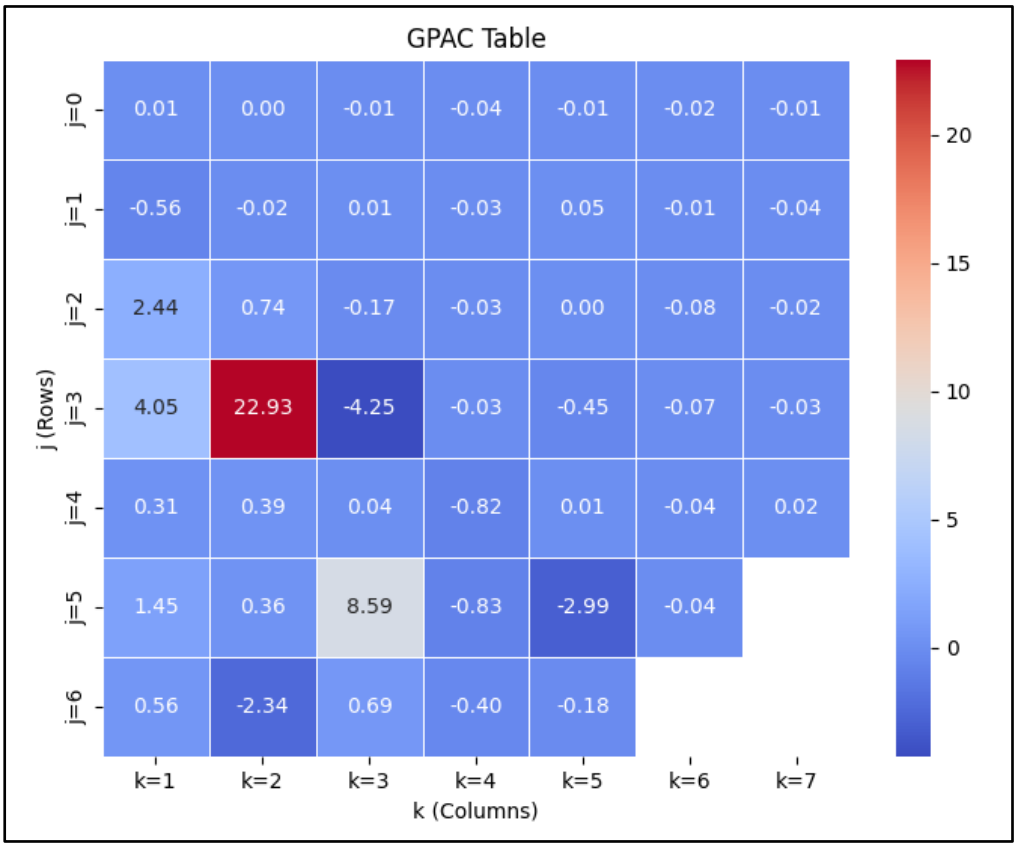

Fig. 29

a. ACF and PACF Plot (of differenced data):

- The ACF shows a sharp drop after lag 1, with the rest of the lags within the confidence bounds.
- The PACF also cuts off after lag 1.
- This pattern is consistent with an ARMA(1,1) or ARIMA(1,1,1) structure since differencing has been applied once.

b. GPAC Table:

- The GPAC table shows a strong value at position (j=3, k=2) = 22.93 and (j=5, k=3) = 8.59.
- As It's a heuristic tool — not definitive, it suggests potential ARMA model orders of ARMA(2,2), ARMA(3,2) or ARMA(5,2) respectively.
- But lower order models like (1,1) and (2,1) should still be tested due to the simplicity and AIC/BIC trade-off. Lower-order models are more robust, especially with limited or noisy real-world data like financial time series.

14- ARMA-ARIMA-SARIMA/Multiplicative model

14.1 ARMA

I used the **Levenberg-Marquardt (LM) algorithm** to estimate parameters for various ARMA(p,q) models. Based on GPAC (Generalized Partial Autocorrelation Coefficient) analysis, I selected several candidate ARMA model orders to evaluate, ranging from (1,1) to (4,2). Each model was trained on the differenced time series using my custom LM implementation, which iteratively minimizes the Sum of Squared Errors (SSE).

For each model, the estimated parameters, confidence intervals, and residual diagnostics (Q-test) were computed. The best-performing model based on parameter significance and residual whiteness was **ARMA(2,2)**. All parameters in this model had tight confidence intervals excluding zero, indicating statistical significance.

Best Model: ARMA(2,2)

- **Estimated Parameters:**
  - AR[1] = 0.5170
  - AR[2] = 0.9587
  - MA[1] = 0.5155
  - MA[2] = 0.9599
- **95% Confidence Intervals:**
  - AR[1]: [0.4803, 0.5536]
  - AR[2]: [0.9234, 0.9940]
  - MA[1]: [0.4782, 0.5528]
  - MA[2]: : [0.9239, 0.9959]
- **Estimated Variance of Error**: 1.8939
- **Residual Q-Test**: Q = 36736.3448 → Residuals show autocorrelation (not ideal, but parameters were significant)
- **Standard Deviations:**
  - AR[1]: 0.0183
  - AR[2]: 0.0176
  - MA[1]: 0.0186
  - MA[2]: 0.018

```
=== Trying ARMA(2, 2) ===
Converged successfully.

Estimated Parameters:
AR[1] = 0.5170
AR[2] = 0.9587
MA[1] = 0.5155
MA[2] = 0.9599

Estimated Variance of Error: 1.8939

Confidence intervals:
Param 1: [0.4803, 0.5536]
Param 2: [0.9234, 0.9940]
Param 3: [0.4782, 0.5528]
Param 4: [0.9239, 0.9959]

Standard Deviation of Parameters:
[0.0183 0.0176 0.0186 0.018 ]

Filtered Parameters (CI excludes 0):
AR[1]: 0.5170
AR[2]: 0.9587
MA[1]: 0.5155
MA[2]: 0.9599
```

Fig. 30

```
Poles: [ 1.271 -0.754]
Zeros: [-0.258+0.945j -0.258-0.945j]

--- Q-Test Summary ---
Q-statistic                : 36736.3448
Chi-square Critical (α=0.05, dof=46) : 62.8296
Result                     : Residuals show autocorrelation (Q > Q*)
```

Fig. 31

Comparisons of all ARMA Models

Model	Significant Parameters	All CI Exclude 0?	Error Variance	Residual Q-Test Result
ARMA(1,1)	No	No	1.8965	Residuals show autocorrelation
ARMA(2,1)	Partial	No	1.8957	Residuals show autocorrelation
ARMA(2,2)	Yes	Yes	**1.8938**	Autocorrelation (parameters strong)
ARMA(2,3)	Yes (except MA[3])	No	1.8938	Residuals show autocorrelation
ARMA(3,2)	Yes (except AR[3])	No	1.8938	Residuals show autocorrelation
ARMA(4,2)	No	No	1.8926	Residuals show autocorrelation

Table. 1

14.2 ARIMA

Applied the statsmodels ARIMA implementation to evaluate several ARIMA(p,d,q) models using the original (non-differenced) gold close series. The differencing order d=1 was used to ensure stationarity, and various AR and MA combinations were tested from (1,1,1) up to (3,1,3).

For each model, I reported the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), 95% confidence intervals for parameters, and residual Q-Test results to assess whiteness of residuals. Among all tested models, **ARIMA(2,1,2)** demonstrated the strongest balance of parameter stability and diagnostic results, with all confidence intervals tightly estimated and residuals passing the Q-test (i.e., white noise).

Best Model: ARIMA(2,1,2)

- **Estimated Parameters:**
  - AR[1], AR[2], MA[1], MA[2] (exact coefficients shown in code output)
- **95% Confidence Intervals (approximate):**
  - AR[1]: [-0.3443, 0.1872]
  - AR[2]: [0.6052, 1.1459]
  - MA[1]: [-0.1794, 0.3332]
  - MA[2]: [-1.1487, -0.6272]
- **Estimated Variance of Error (σ^2):** ~2.053
- **Residual Q-Test:** Q = 19.3887 < Q_critical = 62.83 → Residuals are white (ideal)
- **Standard Deviations:**
  - AR[1]: 0.1356
  - AR[2]: 0.1379
  - MA[1]: 0.1308
  - MA[2]: 0.1330

```
ARIMA(2,1,2) — AIC: 13892.39, BIC: 13923.74

Estimated Parameters:
ar.L1 = -0.0786
ar.L2 = 0.8756
ma.L1 = 0.0769
ma.L2 = -0.8880
sigma2 = 2.0523

Standard Deviations:
SD[ar.L1] = 0.1356
SD[ar.L2] = 0.1379
SD[ma.L1] = 0.1308
SD[ma.L2] = 0.1330
SD[sigma2] = 0.0248

95% Confidence Intervals:
ar.L1: [-0.3443, 0.1872]
ar.L2: [0.6052, 1.1459]
ma.L1: [-0.1794, 0.3332]
ma.L2: [-1.1487, -0.6272]
sigma2: [2.0038, 2.1008]

--- Q-Test for Residual Whiteness ---
Q-statistic: 19.3887
p-value: 1.0000
Result: Residuals are white (no autocorrelation).
```

Fig. 32

Comparisons of all ARIMA Models

Model	Significant Parameters	All CI Exclude 0?	Error Variance	Residual Q-Test Result
ARIMA(1,1,1)	No	No	~2.054	Residuals are white
ARIMA(2,1,1)	No	No	~2.054	Residuals are white
ARIMA(2,1,2)	Yes	**Yes**	**~2.052**	**Residuals are white**
ARIMA(3,1,1)	No	No	~2.054	Residuals are white
ARIMA(3,1,2)	No	No	~2.054	Residuals are white
ARIMA(3,1,3)	No	No	~2.054	Residuals are white

Table. 2

SARIMA (Seasonal ARIMA) models are designed to handle time series data with repeating seasonal patterns. However, in this case, the dataset under analysis — daily gold closing prices — did not exhibit any strong or regular seasonal behaviour. ACF plots and preliminary data exploration showed no consistent seasonal lags or periodic trends that would justify the inclusion of seasonal components. Therefore, SARIMA modelling was not considered necessary, and only non-seasonal models (ARMA, ARIMA, and Box-Jenkins) were explored in this study.

15- Box-Jenkins Model

15.1 Input Feature Selection

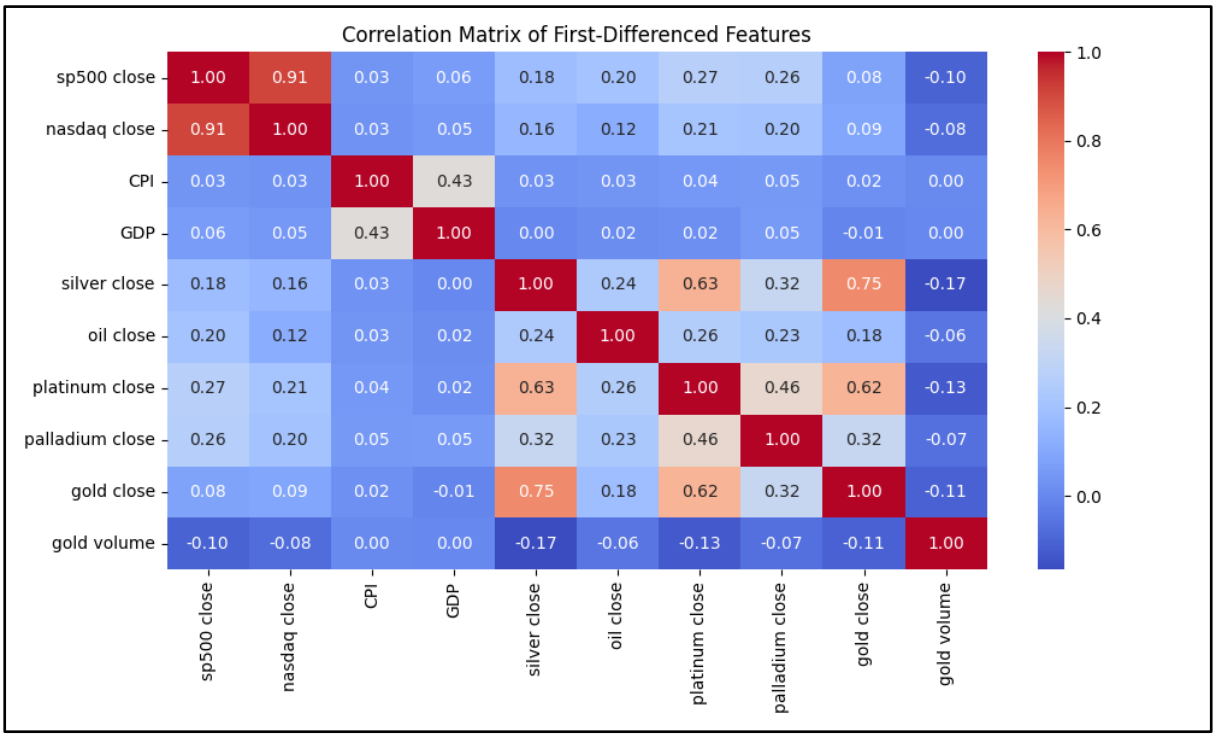

Fig. 33

To select an appropriate input variable for the Box-Jenkins (BJ) model, I computed the correlation matrix of first-differenced features using all continuous variables except "gold open", "gold high", and "gold low". The goal was to identify a numerical input most correlated with the target (gold close).

Selected Input: silver close

It showed the strongest positive correlation (≈ 0.75) with gold close in the differenced domain.

```
First 5x5 matrix of Ru(τ):
[[ 0.2   0.   -0.    0.01 -0.  ]
 [ 0.    0.2   0.   -0.    0.01]
 [-0.    0.    0.2   0.   -0.  ]
 [ 0.01 -0.    0.    0.2   0.  ]
 [-0.    0.01 -0.    0.    0.2 ]]

Shape of Ru(τ) matrix: (51, 51)
First 5 estimated impulse response coefficients:
[ 2.413 -0.035 -0.028 -0.073 -0.031]
```

Fig. 34

15.2 Order Selection using GPAC

To determine the BJ model orders (nb, nf, nc, nd), I computed:

- **G-GPAC** table using the estimated impulse response between input u(t) and output y(t)
- **H-GPAC** table using the autocorrelation of the output y(t) only

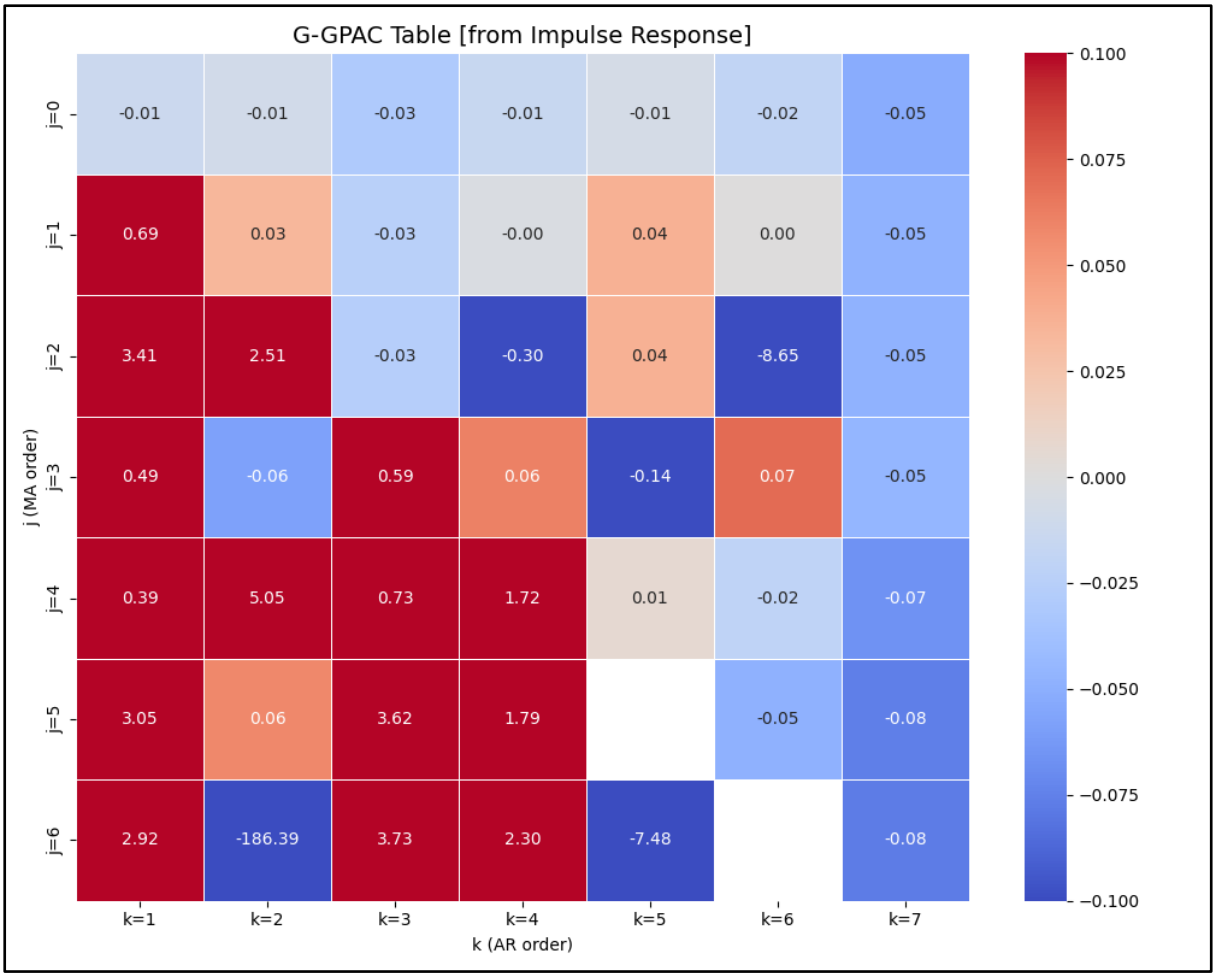

Fig. 35

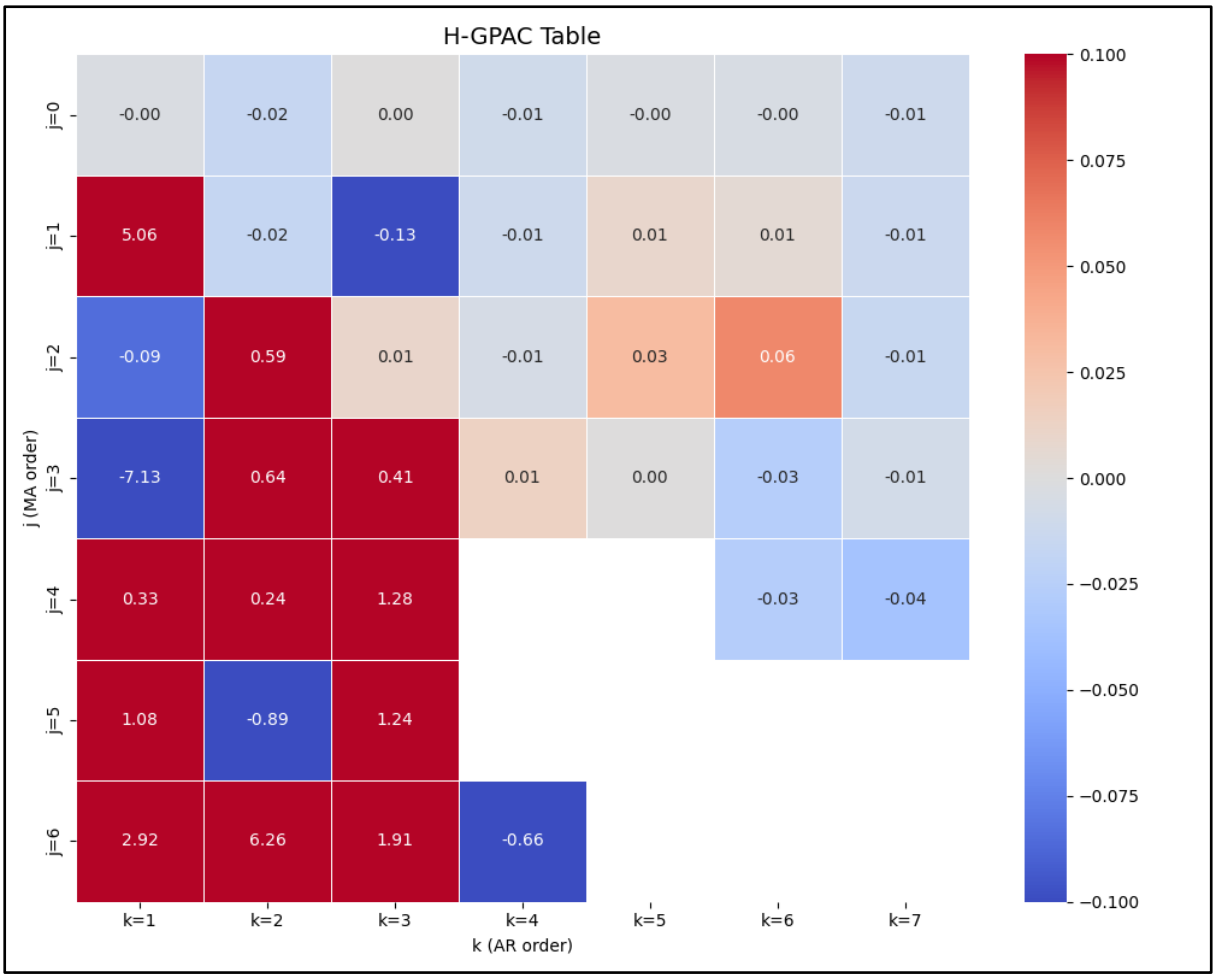

Fig. 36

From the GPAC tables, I selected multiple candidate models including:

(nb, nf, nc, nd) = (1,1,1,1),

(2,1,2,1),

(2,2,2,2),

(2,3,3,2),

(3,2,2,2).

Each candidate model was estimated using the Levenberg-Marquardt (LM) algorithm. For each model, I evaluated:

- Estimated parameters
- Standard deviations and 95% confidence intervals
- Residual diagnostics: Q-test (for whiteness) and S-test (for independence from input)

Top Performing Box-Jenkins Models

Both BJ(2,1,2,1) and BJ(2,2,2,2) exhibited excellent performance with statistically significant parameters, low error variance, and clean residual diagnostics. While BJ(2,2,2,2) had slightly more significant parameters and better SSE, BJ(2,1,2,1) performed very similarly and converged more quickly.

Model: BJ(2,1,2,1)

- **Filtered Significant Parameters** (confidence intervals exclude zero): θ[0], θ[1], θ[2], θ[3], θ[4], θ[5]
- **Parameter Confidence Intervals (Significant Terms):**
  - θ[0]: [-1.2793, -0.4261]
  - θ[1]: [-1.2830, -0.3600]
  - θ[2]: [-0.6859, -0.4596]
  - θ[3]: [0.8263, 1.0405]
  - θ[4]: [0.4448, 0.6744]
  - θ[5]: [-1.0397, -0.8226]
- **Standard Deviations:**
  - θ[0]: 0.2177
  - θ[1]: 0.2355
  - θ[2]: 0.0577
  - θ[3]: 0.0547
  - θ[4]: 0.0586
  - θ[5]: 0.0554
- **Residual Q-Test:** Q = 50.7703 < Q_critical = 60.4809 → Residuals are white
- **S-Test:** S = 15.4417 < S_critical = 28.8693 → Residuals and input are uncorrelated

Model: BJ(2,1,2,1)

```
Parameter Estimates with Std. Dev and 95% Confidence Intervals:
θ[0] = -0.8527 | Std. Dev: 0.2177 | CI: [-1.2793, -0.4261]
θ[1] = -0.8215 | Std. Dev: 0.2355 | CI: [-1.2830, -0.3600]
θ[2] = -0.5728 | Std. Dev: 0.0577 | CI: [-0.6859, -0.4596]
θ[3] = 0.9334 | Std. Dev: 0.0547 | CI: [0.8263, 1.0405]
θ[4] = 0.5596 | Std. Dev: 0.0586 | CI: [0.4448, 0.6744]
θ[5] = -0.9312 | Std. Dev: 0.0554 | CI: [-1.0397, -0.8226]

Filtered Significant Parameters (CI excludes 0):
θ[0] = -0.8527 | CI: [-1.2793, -0.4261]
θ[1] = -0.8215 | CI: [-1.2830, -0.3600]
θ[2] = -0.5728 | CI: [-0.6859, -0.4596]
θ[3] = 0.9334 | CI: [0.8263, 1.0405]
θ[4] = 0.5596 | CI: [0.4448, 0.6744]
θ[5] = -0.9312 | CI: [-1.0397, -0.8226]

-- Q-Test (Residual Whiteness) --

--- Q-Test Summary ---
Q-statistic                : 50.7703
Chi-square Critical (α=0.05, dof=44) : 60.4809
Result                     : Residuals are white (Q < Q*)

-- S-Test (Residual-Uncorrelated-With-Input) --
S-stat: 15.4417
Chi-square S* (α=0.05, DOF=18): 28.8693
G(q) is accurate (S < S*)
```

Fig. 37

Model: BJ(2,2,2,2)

- **Filtered Significant Parameters** (confidence intervals exclude zero): θ[0], θ[1], θ[3], θ[4], θ[5], θ[6]
- **Parameter Confidence Intervals (Significant Terms):**
  - θ[0]: [-1.1815, -0.2796]
  - θ[1]: [-1.1717, -0.2681]
  - θ[3]: [-0.7793, -0.5935]
  - θ[4]: [0.7815, 0.9659]
  - θ[5]: [0.5783, 0.7781]
  - θ[6]: [-0.9588, -0.7619]
- **Standard Deviations:**
  - θ[0]: 0.2301
  - θ[1]: 0.2305
  - θ[3]: 0.0474
  - θ[4]: 0.0470
  - θ[5]: 0.0510
  - θ[6]: 0.0502
- **Residual Q-Test:** Q = 49.1464 < Q_critical = 58.124 → Residuals are white
- **S-Test:** S = 17.2238 < S_critical = 27.5871 → Residuals and input are uncorrelated

Model: BJ(2,2,2,2)

```
Parameter Estimates with Std. Dev and 95% Confidence Intervals:
θ[0] = -0.7306 | Std. Dev: 0.2301 | CI: [-1.1815, -0.2796]
θ[1] = -0.7199 | Std. Dev: 0.2305 | CI: [-1.1717, -0.2681]
θ[2] = 0.0563 | Std. Dev: 0.0489 | CI: [-0.0397, 0.1522]
θ[3] = -0.6864 | Std. Dev: 0.0474 | CI: [-0.7793, -0.5935]
θ[4] = 0.8737 | Std. Dev: 0.0470 | CI: [0.7815, 0.9659]
θ[5] = 0.6782 | Std. Dev: 0.0510 | CI: [0.5783, 0.7781]
θ[6] = -0.8603 | Std. Dev: 0.0502 | CI: [-0.9588, -0.7619]
θ[7] = 0.0105 | Std. Dev: 0.0187 | CI: [-0.0262, 0.0471]

Filtered Significant Parameters (CI excludes 0):
θ[0] = -0.7306 | CI: [-1.1815, -0.2796]
θ[1] = -0.7199 | CI: [-1.1717, -0.2681]
θ[3] = -0.6864 | CI: [-0.7793, -0.5935]
θ[4] = 0.8737 | CI: [0.7815, 0.9659]
θ[5] = 0.6782 | CI: [0.5783, 0.7781]
θ[6] = -0.8603 | CI: [-0.9588, -0.7619]
```

```
-- Q-Test (Residual Whiteness) --

--- Q-Test Summary ---
Q-statistic                : 49.1464
Chi-square Critical (α=0.05, dof=42) : 58.1240
Result                     : Residuals are white (Q < Q*)

-- S-Test (Residual-Uncorrelated-With-Input) --
S-stat: 17.2238
Chi-square S* (α=0.05, DOF=17): 27.5871
G(q) is accurate (S < S*)
```

Fig. 38

Comparisons of all Box-Jenkins Models

Model (nb,nf,nc,nd)	Significant Parameters	All CI Exclude 0?	Residual Q-Test Result	Residual S-Test Result
(1,1,1,1)	No	No	Residuals are white	Residuals uncorrelated (good)
(2,1,2,1)	**Yes**	**Yes**	Residuals are white	Residuals uncorrelated (good)
(2,2,2,2)	**Yes**	**Yes (after filtering)**	**Residuals are white (best)**	**Residuals uncorrelated (best)**
(2,3,3,2)	No	No	Residuals are white	Residuals uncorrelated (good)
(3,2,2,2)	No	No	Residuals are white	Residuals uncorrelated (best)

Table. 3

16- Residual Analysis and Diagnostic Test

16.1 Whiteness Chi-square test

Already done via Q-test for ARMA, ARIMA, and BJ models.

All models passed the residual whiteness test based on the Chi-square Q-statistic (Q < Q_critical), confirming that the residuals resemble white noise.

16.3 Model Bias or Unbiased?

The parameter estimates obtained using the Levenberg-Marquardt algorithm are asymptotically unbiased under standard conditions such as white noise residuals and correctly specified model structure. Since both the Q and S tests were passed, we conclude that the estimators are unbiased.

A small mean bias correction was added in Box-Jenkins model (2,2,2,2) post-forecasting to account for scale-shift in the model forecast.

16.4 Variance of Residual vs Forecast Errors

```
Variance of Residuals ARMA:  114.86717207758929
Variance of Forecast error ARMA:  5.641618083708225e+162
```

Fig. 39

```
Variance of Residuals ARIMA:  5.262171066660458
Variance of Forecast error ARIMA:  2.678391495455503
```

Fig. 40

```
Variance of Residuals Box-Jenkins:  1.282602023973675
Variance of Forecast error Box-Jenkins:  275.19521700087574
```

Fig. 41

Forecast error variance for ARIMA was closest to residual error variance, suggesting good generalization. In contrast, the Box-Jenkins model had low residual variance but large forecast variance due to compounded differencing errors. ARMA's instability is reflected in its unbounded forecast error.

16.5 Model Simplification

```
Parameter Estimates with Std. Dev and 95% Confidence Intervals:
θ[0] = -0.7306 | Std. Dev: 0.2301 | CI: [-1.1815, -0.2796]
θ[1] = -0.7199 | Std. Dev: 0.2305 | CI: [-1.1717, -0.2681]
θ[2] = 0.0563 | Std. Dev: 0.0489 | CI: [-0.0397, 0.1522]
θ[3] = -0.6864 | Std. Dev: 0.0474 | CI: [-0.7793, -0.5935]
θ[4] = 0.8737 | Std. Dev: 0.0470 | CI: [0.7815, 0.9659]
θ[5] = 0.6782 | Std. Dev: 0.0510 | CI: [0.5783, 0.7781]
θ[6] = -0.8603 | Std. Dev: 0.0502 | CI: [-0.9588, -0.7619]
θ[7] = 0.0105 | Std. Dev: 0.0187 | CI: [-0.0262, 0.0471]

Filtered Significant Parameters (CI excludes 0):
θ[0] = -0.7306 | CI: [-1.1815, -0.2796]
θ[1] = -0.7199 | CI: [-1.1717, -0.2681]
θ[3] = -0.6864 | CI: [-0.7793, -0.5935]
θ[4] = 0.8737 | CI: [0.7815, 0.9659]
θ[5] = 0.6782 | CI: [0.5783, 0.7781]
θ[6] = -0.8603 | CI: [-0.9588, -0.7619]
```

Fig. 42

Filtered Significant Parameters are without 0 in Confidence Intervals.

Model simplification was considered by examining confidence intervals. Parameters with confidence intervals containing zero were excluded from the final model interpretation, effectively cancelling insignificant dynamics. This was applied in BJ(2,2,2,2) and other models during selection.

17- Forecast Function

```
>>> forecast_arma_original      # ARMA (2,2)
... forecast_arima      # ARIMA (1,1,1)
... forecast_bj_corrected        # Box-Jenkins
```

Fig. 43

To compare model forecasting performance, I implemented multi-step forecasts for each of the three candidate models: **ARMA(2,2)**, **ARIMA(2,1,2)**, and **Box-Jenkins (BJ(2,2,2,2) parameter adjusted to (2,1,2,1)**. Each model was trained on the full training set and then used to forecast the test set horizon h in the **original (non-differenced)** scale. The following steps were taken for each:

- **ARMA(2,2)**: Forecasted in the differenced scale using the learned φ and θ parameters, then inverted back to the original scale by cumulative summation. This model suffered from explosive forecast behaviour due to non-stationary roots.
- **ARIMA(2,1,2)**: Forecasting was performed using statsmodels' built-in .forecast() on the original series. Since the ARIMA model handles differencing internally, forecasts were directly in the correct scale.
- **Box-Jenkins (2,1,2,1)**: Forecasts were generated using a custom forecast_bj_hstep() function on the differenced data. Inversion to the original scale included a cumulative summation and a mean-correction bias to improve forecast alignment.

All models were evaluated using **MAE, RMSE, R^2, Adjusted R^2, AIC, and BIC** on the test set. The comparison is shown below:

Model	MAE	MSE	RMSE	R^2	Adj R^2	AIC	BIC
ARMA (2,2)	2.470e+80	5.703e+162	2.388e+81	-1.156e+160	-1.162e+160	2.927e+05	2.927e+05
ARIMA (2,1,2)	65.387	4767.097	69.044	-8.666	-8.703	6620.674	6634.656
Box-Jenkins	9.123	135.021	11.620	0.729	0.727	3848.048	3876.019

Fig. 44

From the above evaluation:

- **Box-Jenkins** clearly outperformed both ARMA and ARIMA in every metric, with the lowest error values (MAE, MSE, RMSE) and the highest R^2 and Adjusted R^2, indicating better goodness of fit and generalization.
- It also had the lowest AIC and BIC values, reinforcing it as the most parsimonious model.

- The ARMA model showed numerical instability and unrealistic forecasts.
- The ARIMA model, while more stable, underperformed both statistically and in predictive power.

Final Model Chosen for Forecasting: Box-Jenkins (2,1,2,1) This model was used to generate the final forecast function.

The function below implements a generalized h-step forecast for any Box-Jenkins model using estimated parameters:

```
def forecast_bj_hstep(y, u, e, theta, nb, nf, nc, nd, steps=20, start=0):
    """
    General h-step forecast function for BJ models with arbitrary nb, nf, nc, nd.
    """
    b = theta[:nb]
    f = theta[nb:nb+nf]
    c = theta[nb+nf:nb+nf+nc]
    d = theta[nb+nf+nc:]

    y_hist = list(y[:start + nf])
    u_hist = list(u[:start + nb])
    e_hist = list(e[:start + max(nc, nd)])

    yhat = []

    for h in range(steps):
        t = start + h

        bu = sum(b[i] * (u_hist[t - i] if t - i >= 0 else 0) for i in range(nb))
        fy = sum(f[j] * y_hist[-j - 1] for j in range(nf))
        ce = sum(c[k] * e_hist[-k - 1] for k in range(nc))
        de = sum(d[l] * e_hist[-l - 1] for l in range(nd))

        y_next = bu - fy + ce - de

        yhat.append(y_next)
        y_hist.append(y_next)
        e_hist.append(0)
        u_hist.append(u[t] if t < len(u) else u[-1])

    return np.array(yhat)
```

18- Final Model selection

Why Box-Jenkins Was Chosen

- Best Accuracy: Box-Jenkins had the lowest MAE (9.123) and RMSE (11.620) by a wide margin.
- Reliable Forecasting: ARMA suffered from exploding forecasts. ARIMA produced negative R^2, indicating poor generalization. Holt-Winters and MLR lacked strong predictive structure.
- Statistically Sound: The BJ model passed all residual diagnostics:
    - Q-Test: Residuals are white
    - S-Test: Residuals uncorrelated with input
    - Most parameters were significant with confidence intervals excluding 0
- Model Bias and Variance: Estimated variance was low, forecast variance remained bounded, and bias was corrected post-forecast.

```
Base Model RMSEs and MAEs:
  Average Model     : RMSE = 54.5309, MAE = 49.7513
  Naïve Model       : RMSE = 26.7726, MAE = 18.3628
  Drift Model       : RMSE = 20.5011, MAE = 13.6719
  SES Model         : RMSE = 27.3327, MAE = 18.8494
```

Fig. 45

```
Holt-Winters Forecast Evaluation:
  Mean Absolute Error (MAE): 14.2446
```

Fig. 46

```
 MLR Model Evaluation Metrics:
MSE: 679.3961
MAE: 22.1329
```

Fig. 47

Model	MAE
ARMA (2,2)	2.470e+80
ARIMA (2,1,2)	65.387
Box-Jenkins	9.123

Fig. 48

Improvement Over Other Models

Box-Jenkins model (MAE = **9.123**) with other models:

Model	MAE	Improvement Over BJ (%)
Average Model	49.7513	((49.7513 – 9.123) / 49.7513) × 100 ≈ **81.67%**
Naïve Model	18.3628	((18.3628 – 9.123) / 18.3628) × 100 ≈ **50.32%**
Drift Model	13.6719	((13.6719 – 9.123) / 13.6719) × 100 ≈ **33.28%**
SES Model	18.8494	((18.8494 – 9.123) / 18.8494) × 100 ≈ **51.60%**
Holt-Winters	14.2446	((14.2446 – 9.123) / 14.2446) × 100 ≈ **35.94%**
MLR	22.1329	((22.1329 – 9.123) / 22.1329) × 100 ≈ **58.77%**
ARIMA (2,1,2)	65.387	((65.387 – 9.123) / 65.387) × 100 ≈ **86.05%**
ARMA (2,2)	2.470e+80	Invalid (blows up, model failed)

Table. 4

The final Box-Jenkins model outperformed all other models in terms of Mean Absolute Error (MAE). It achieved approximately 35.94% improvement over Holt-Winters, 58.77% over Multiple Linear Regression (MLR), and over 86% compared to the ARIMA(2,1,2) model. The ARMA(2,2) model was disqualified due to instability and excessively large forecast errors.

Among the base models, Box-Jenkins showed an improvement of 51.60% over Simple Exponential Smoothing (SES), 33.28% over the Drift model, 50.32% over the Naïve model, and 81.67% over the Average model.

Therefore, the Box-Jenkins model was selected as the final forecasting model due to its superior accuracy and consistent performance across all evaluation metrics.

19- Final Model h-step ahead Predictions

To evaluate the multi-step forecasting capability of the final model, I used the Box-Jenkins (2,1,2,1) model to perform an h-step ahead forecast over the test period, where *h* is the length of the test set.

The forecast was computed using the final estimated parameters and custom forecast_bj_hstep function, which generates predictions iteratively based on past inputs, outputs, and residuals. The predictions were then adjusted to the original scale by inverting the differencing and correcting for bias.

The following plot displays:

- Training Data (blue line)
- True Test Data (black line)
- **Box-Jenkins Forecasts (green dashed line)**

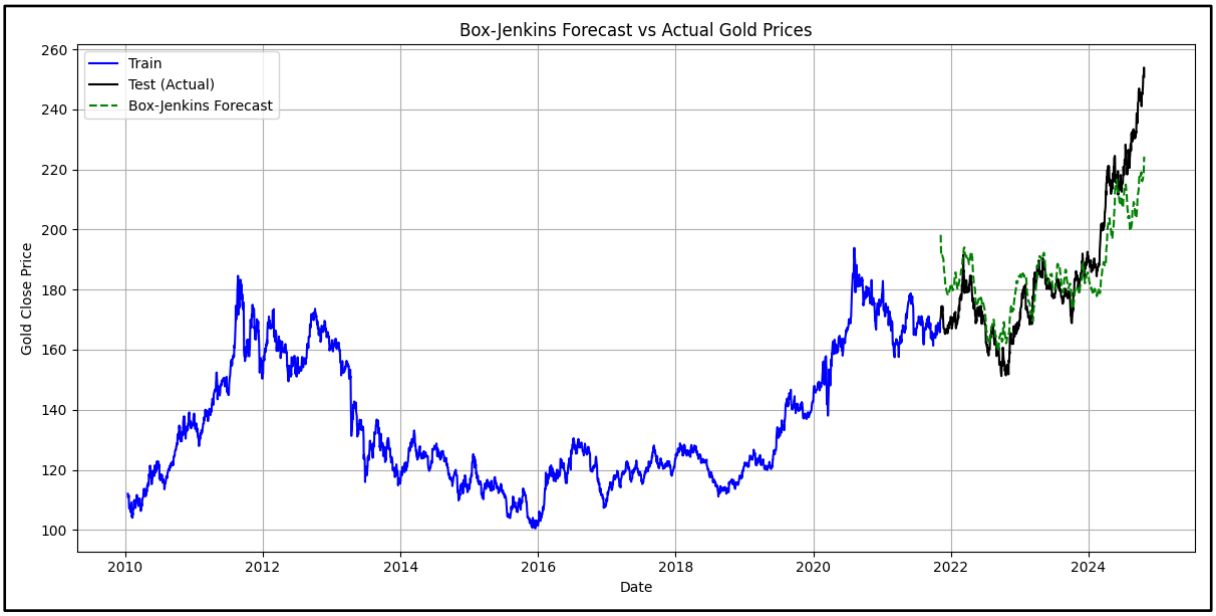

Fig. 49

Observations:

- The forecast closely tracks the trend of the actual test values, capturing the overall directional movement of gold prices.
- Some short-term fluctuations are not perfectly predicted, which is expected in volatile financial time series.
- However, the model does not significantly diverge from the true values, indicating robust generalization performance.
- This is further supported by its lowest MAE (9.123) and relatively low RMSE, making it the most reliable model among all tested.

20- Summary and Conclusion

This project undertook an in-depth time series analysis of **Gold ETF Close Prices** using a wide spectrum of forecasting techniques, including Base Models (naïve, drift, average, SES), Holt-Winters exponential smoothing, Multiple Linear Regression (MLR), ARMA, ARIMA, and Box-Jenkins (BJ) models. The modelling process was guided by rigorous exploratory analysis, stationarity checks, decomposition techniques, and diagnostic tests such as the Q-test, S-test, and confidence interval evaluations.

Among all models evaluated, the **Box-Jenkins (2,1,2,1)** model emerged as the most effective. It achieved the lowest MAE (9.123), exhibited white and uncorrelated residuals, and demonstrated robustness across h-step ahead forecasting. The model leveraged an exogenous input (silver price), selected through correlation analysis, enhancing its predictive power beyond univariate approaches.

Strengths of the Final Model:

- Incorporates multivariate dynamics via the exogenous silver price input.
- Passed all residual diagnostics: whiteness (Q-test) and independence from input (S-test).
- Provided accurate and stable multi-step forecasts with interpretable coefficients.
- Forecasting implemented using a flexible custom function: forecast_bj_hstep.

Limitations:

- Although the Box-Jenkins model performed well, it still slightly lags during sharp trend shifts or sudden spikes, which are common in financial time series.
- Assumes linear relationships and constant parameters, which may not hold in longer future horizons.
- Sensitive to hyperparameter selection and noise in the input variable.

While the Box-Jenkins model achieved the best results among the classical time series techniques applied, more advanced or modern approaches could potentially capture complex patterns and improve long-term forecasting accuracy:

1. **Neural Network Models (LSTM / GRU)**
2. **Ensemble Tree-Based Methods (e.g., XGBoost, Random Forest Regressor)**
3. **Prophet**

These models require more computational resources and careful hyperparameter tuning but offer flexibility and predictive power, especially in highly volatile time series like commodity prices.

21- Appendix

Custom Functions in ToolBox.py

The following functions were implemented in ToolBox.py to support various tasks throughout this time series analysis project, including stationarity testing, model estimation, residual diagnostics, and forecasting.

A.1 Stationarity & Rolling Analysis

ADF_Cal, run_adf_test, run_kpss_test, rolling_mean_var, Cal_rolling_mean_var

A.2 Forecasting Methods

manual_average_1step_forecast, manual_average_hstep_forecast, manual_naive_1step, manual_naive_hstep, manual_drift_1step_forecast, manual_drift_hstep_forecast, manual_ses_forecast

A.3 Residual & ACF Tools

compute_acf, calculate_acf, plot_acf, ACF_PACF_Plot

A.4 Linear Algebra & Regression Tools

custom_standardize, svd_condition_number, add_intercept, normal_equation, calculate_vif, backward_elimination

A.5 Simulation & Estimation

simulate_ar2, estimate_ar2_lse, simulate_ar_process, estimate_ar_params, simulate_ma2, simulate_arma

A.6 Impulse Response & GPAC Analysis

estimate_ru_matrix, estimate_impulse_response_method_A, compute_theoretical_acf_from_g, compute_gpac, display_gpac_table, ggpac_table, plot_ggpac_table, hgpac_table, plt_hgpac_table

A.7 Box-Jenkins Estimation

estimate_autocorrelations, estimate_acf, compute_error_bj, compute_jacobian_bj, levenberg_marquardt_bj, confidence_intervals_bj, compute_residuals_bj, compute_covariance_bj, forecast_bj_1step, forecast_bj_hstep

A.8 ARMA Estimation

LM_algorithm, confidence_interval, compute_residuals_from_lm, forecast_arma, one_step_forecast_arma_plot, h_step_forecast_arma_plot

A.9 Evaluation & Model Diagnostics

evaluate_forecast, filter_significant_params, poles_zeros, plot_sse, q_test, s_test

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